

SHUFAN SHAHI

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EDUCATION

Islamic University of Technology

2022–2026

B.Sc. in Computer Science and Engineering

CGPA: 3.58

Coursework: Algorithms, Machine Learning, Deep Learning, Databases, and Networks.

Chuadanga Govt. College

2021

Higher Secondary Certificate (HSC)

GPA: 5.00/5.00

EXPERIENCE

Student Researcher

2026–Present

Elite Research Lab

- Support research projects through literature reviews, experiments, and analysis.

Intern

2025

Battery Low Interactive Ltd.

- Learned Flutter development, REST API integration, and applications of fine-tuned ML models.

Researcher

2025–2026

IUT Computer Vision Lab

- Researched multimodal AI methods to improve integration and mitigate modality imbalance.

Senior Executive

2025–2026

Project Aero

- Contributed to drone R&D by applying AI-based object tracking for international UAV competitions.

PROJECTS

DocuMind - Document Q&A | Python, LangChain, LlamaIndex, ChromaDB, Ollama

🐙 GitHub

- Built a local RAG system for documents with cited answers and hallucination guardrails.
- Compared 4 chunking methods and 2 embedding models; added hybrid BM25 and reranking.

Benhallu - Bengali Hallucination Detection | Python, RAG, BanglaBERT, scikit-learn

🐙 GitHub

- Built a retrieval-augmented pipeline to verify Bengali answers against supplied context or a Bengali QA knowledge base.
- Created a hallucination dataset and used fine-tuned BanglaBERT with answer-comparison features to classify responses.

Bengali Speaker Diarization - Speaker Turn Detection | PyTorch, Silero VAD, ECAPA-TDNN

🐙 GitHub

- Built a pipeline using neural voice activity detection, speaker embeddings, and agglomerative clustering to label timestamped speaker turns.
- Estimated speaker counts with silhouette scores and exported diarization results to CSV.

Drone Vision - Aerial Tracking | PyTorch, YOLOv11X, ByteTrack

🐙 GitHub

- Fine-tuned YOLOv11X on VisDrone and used ByteTrack for human counting and multi-object tracking.

BD Traffic Monitoring System - Traffic Analytics | PyTorch, YOLO11L, Flask, Docker

🐙 GitHub 🖱 Live

- Fine-tuned YOLO on Bangladesh traffic data to detect local vehicle types.

Biztrack - Business Analytics | *Next.js, Node.js, PostgreSQL*

[GitHub](#) [Demo](#)

- Built AI modules for real-time finance tracking, risk alerts, customer segmentation, and forecasting.

Postfolio - Professional Networking | *Spring Boot, Next.js, Docker, RabbitMQ, Stripe*

[GitHub](#) [Demo](#)

- Integrates social networking, portfolio and CV creation, job search, and AI interview preparation.

Redrick Routinson - Campus Automation | *Python, React, Flask, Firebase*

[GitHub](#) [Demo](#)

- Built GA scheduling, QR exam attendance, class swapping, events, forums, and lost-and-found.

RESEARCH EXPERIENCES

EthMuSAM: Mutual Scoring with SAM for Zero-Shot Anomaly Segmentation

2026

- Combined DINOv2 patch embeddings with SAM-based mutual scoring to localize zero-shot anomalies.

Prototype-Guided Adaptive Feature Learning for Multimodal Emotion Recognition in Conversation (MERC)

2026

- Used adaptive weighting and modality prototypes to improve multimodal emotion recognition.

Rethinking Visual Features in Graph-Based Multimodal Emotion Recognition in Conversations (MERC): From Low-Level Pixels to Semantic Representations

2026

- Used semantic visual representations to improve graph-based multimodal emotion recognition.

A Cross-Lingual Information Retrieval System for Bangla-English News Corpora

2025

- Developed a hybrid CLIR system using lexical, fuzzy, and embedding-based retrieval.
- Handled cross-language and code-switched queries with semantic ranking.

Weakly Supervised Named Entity Recognition via Sparse-CHMM and BERT

2025

- Aggregated noisy labeling functions without ground-truth annotations.
- Modeled sequence dependencies using Sparse-CHMM with contextual embeddings.

TECHNICAL SKILLS

Programming Languages: Python, C++, Java

Data and Machine Learning: PyTorch, TensorFlow, scikit-learn, NumPy, Pandas

Databases: MySQL, Firebase, PostgreSQL

Web Development: React, Tailwind CSS, Next.js, Spring Boot

Mobile Development: React Native, Flutter

Visualization: Matplotlib, Seaborn

Development Tools: Git, Linux

ACHIEVEMENTS, AWARDS AND SCHOLARSHIPS

- **5th Place, VAND4 International Competition** — Zero-Shot Industrial Track 2026
- **First Runner-up, IUT Coderush Hackathon** — Rapid team application development 2025
- **7th Place, BAIC Datathon** — Data preprocessing, visualization, and modeling 2025

REFERENCE

Shahriar Ivan

Assistant Professor, Department of CSE, IUT

Specialization: Machine Learning, Deep Learning, Computer Vision

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